

BIT MANIPULATION

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The first step to achieving your goal, is to take a moment to respect your goal. Know what it means to you to achieve it.

DWAYNE JOHNSON



Good
Evening

Today's content

01. Check bit is set or not
02. Count set bits
03. Power of left shift
 - (a) Set i^{th} bit
 - (b) Flip i^{th} bit
04. Negative number
05. Ranges

Revision

Right shift operator

$$A = 20 \quad \underline{0} \quad \underline{0} \quad \underline{0} \quad \underline{1} \quad \underline{0} \quad \underline{1} \quad \underline{0} \quad \underline{0} = 20$$

$$A \gg 1 \quad \underline{0} \quad \underline{0} \quad \underline{0} \quad \underline{0} \quad \underline{1} \quad \underline{0} \quad \underline{1} \quad \underline{0} = 10 = \frac{A}{2^1}$$

$$A \gg 2 \quad \underline{0} \quad \underline{0} \quad \underline{0} \quad \underline{0} \quad \underline{0} \quad \underline{1} \quad \underline{0} \quad \underline{1} = 5 = \frac{A}{2^2}$$

No overflow

$$a \gg n = \frac{a}{2^n}$$

Quiz : $N \& 1$ $\begin{cases} 1 & \text{if } N \text{ is odd (last bit is set)} \\ 0 & \text{if } N \text{ is even (last bit is unset)} \end{cases}$

Q1. Given N & i , check if i^{th} bit in N is set or unset.

Ex: $N=21$ $\begin{array}{cccccc} 5 & 4 & 3 & 2 & 1 & 0 \\ \underline{0} & \underline{1} & \underline{0} & \underline{1} & \underline{0} & \underline{1} \end{array}$

$i=2$

→ return true

$N=34$ $\begin{array}{cccccc} 5 & 4 & 3 & 2 & 1 & 0 \\ \underline{1} & \underline{0} & \underline{0} & \underline{0} & \underline{1} & \underline{0} \end{array}$

$i=3$

→ return false

Idea 1 :- Store binary of N & check if i^{th} bit is set or not
[TODO]

Idea 2 :- Right shift operator with &

$N=10$ $\begin{array}{cccccc} 2^7 & 2^6 & 2^5 & 2^4 & 2^3 & 2^2 & 2^1 & 2^0 & i^{\text{th}} \\ \underline{0} & \underline{0} & \underline{0} & \underline{0} & \underline{1} & \underline{0} & \underline{1} & \underline{0} & 0 \end{array}$
 $N \gg 1$ $\begin{array}{cccccc} \underline{0} & \underline{0} & \underline{0} & \underline{0} & \underline{1} & \underline{0} & \underline{1} & \underline{1} & 1 \end{array}$

$N \& 1 == 1 \rightarrow 0^{\text{th}}$ bit is set

$(N \gg 1) \& 1 == 1 \rightarrow 1^{\text{th}}$ bit is set

$N=45$ $\begin{array}{cccccc} 2^7 & 2^6 & 2^5 & 2^4 & 2^3 & 2^2 & 2^1 & 2^0 & i^{\text{th}} \\ \underline{0} & \underline{0} & \underline{1} & \underline{0} & \underline{1} & \underline{1} & \underline{0} & \underline{1} & 0 \end{array}$
 $N \gg 2$ $\begin{array}{cccccc} \underline{0} & \underline{0} & \underline{0} & \underline{0} & \underline{1} & \underline{0} & \underline{1} & \underline{1} & 2 \end{array}$

$(N \gg 2) \& 1 == 1 \rightarrow 2^{\text{nd}}$ bit is set

$N=45$ $\begin{array}{cccccc} 2^7 & 2^6 & 2^5 & 2^4 & 2^3 & 2^2 & 2^1 & 2^0 & i^{\text{th}} \\ \underline{0} & \underline{0} & \underline{1} & \underline{0} & \underline{1} & \underline{1} & \underline{0} & \underline{1} & 0 \end{array}$
 $N \gg 3$ $\begin{array}{cccccc} \underline{0} & \underline{0} & \underline{0} & \underline{0} & \underline{0} & \underline{1} & \underline{0} & \underline{1} & 3 \end{array}$

$(N \gg 3) \& 1 == 1 \rightarrow 3^{\text{rd}}$ bit is set

boolean checkbit(int N , int i)

if $((N \gg i) \& 1 == 1)$ {

 return true

}

else {

 return false

}

TC: $O(1)$

SC: $O(1)$

02. Given N , return no. of setbits in N .

Ex: $N=10$ 1 0 1 0 : return 2

$N=27$ 1 1 0 1 1 : return 4

Idea 1 → Iterate on all the bits & check if it's set or not.

4 bytes = 32 bits

```
int countbits (int n)
|
|   int c = 0
|
|   for (i=0; i<32; i++) {
|       |   if (N >> i & 1 == 1) { c = c + 1 }
|       |
|       |   3
|       |
|       |   return c;
|
|   3
```

01. Issue → Only works for int / for long → $i=0; i<64$

→ Unnecessary iteration

Idea 2 → Move each & every bit to 0th index & then check if it is set or not ($n \& 1 == 1$)

$N=45$, $c=0$

i

2^6	2^5	2^4	2^3	2^2	2^1	2^0
0	1	0	1	1	0	1
0	0	1	0	1	1	0
0	0	0	1	0	1	1
0	0	0	0	1	0	1
0	0	0	0	0	1	0
0	0	0	0	0	0	1
0	0	0	0	0	0	0

$c = c + 1 = 1$

not set

$c = c + 1 = 2$

$c = c + 1 = 3$

not set

$c = c + 1 = 4$

$n = 0 \rightarrow \text{stop}$

int countSetBits2 (int n)

int c = 0

while (n > 0)

if (n & 1 == 1)

c = c + 1

n = n >> 1; // n = n/2

return c;

$Tc = O(\log n)$

$Sc = O(1)$

Left shift operator

int $\rightarrow$ 4 bytes $\rightarrow$ 32 bits

// Assumption int $\rightarrow$ 8 bits } Just for the explanation

$A = 45$ 0 0 1 0 1 1 0 1 = 45 = 45×2^0
 $A \ll 1$ x 0 1 0 1 1 0 1 = 90 = 45×2^1
 $A \ll 2$ x 1 0 1 1 0 1 0 = 180 = 45×2^2
 $A \ll 3$ x 0 1 1 0 1 0 0 = 360 = 45×2^3
 overflow $\rightarrow$ ans = 104 & not 360

$a \ll n = a * 2^n$

$7 \ll 3 = 7 * 2^3$

$= 7 * 8 = \underline{\underline{56}}$

$a \ll n = a * 2^n$

$1 \ll n = 2^n$

Power of left shift

$2^5 \ 2^4 \ 2^3 \ 2^2 \ 2^1 \ 2^0$

1: 0 0 0 0 0 1

$1 \ll 3$ 0 0 1 0 0 0

$\hookrightarrow 2^3 = 8$

$2^5 \ 2^4 \ 2^3 \ 2^2 \ 2^1 \ 2^0$

1: 0 0 0 0 0 1

$1 \ll 2$ 0 0 0 1 0 0 $= 2^2 = 4$

<< with OR

$2^7 \ 2^6 \ 2^5 \ 2^4 \ 2^3 \ 2^2 \ 2^1 \ 2^0$

$N = 45$ 0 0 1 0 1 1 0 1 same

or

$(1 \ll 2) =$ 0 0 0 0 0 1 0 0

$N \mid (1 \ll 2) =$ 0 0 1 0 1 1 0 1

$2^7 \ 2^6 \ 2^5 \ 2^4 \ 2^3 \ 2^2 \ 2^1 \ 2^0$

$N = 45$ 0 0 1 0 1 1 0 1

or

$(1 \ll 4) =$ 0 0 0 1 0 0 0 0

$N \mid (1 \ll 4) =$ 0 0 1 1 1 1 0 1 } $N + 2^4$

$2^7 \ 2^6 \ 2^5 \ 2^4 \ 2^3 \ 2^2 \ 2^1 \ 2^0$

$N = 42$ 0 0 1 0 1 0 1 0 same

1

$(1 \ll 3) =$ 0 0 0 0 1 0 0 0

$N \mid (1 \ll 3) =$ 0 0 1 0 1 0 1 0

$2^7 \ 2^6 \ 2^5 \ 2^4 \ 2^3 \ 2^2 \ 2^1 \ 2^0$

$N = 50$ 0 0 1 1 0 0 1 0

1

$(1 \ll 3) =$ 0 0 0 0 1 0 0 0

$N \mid (1 \ll 3) =$ 0 0 1 1 1 0 1 0 } $N + 2^3$

Obs: $N \mid (1 \ll i) :$

if i^{th} bit of N is set $\rightarrow$ ans = same

if i^{th} bit is unset = ans = $N + 2^i$

Q: Set i^{th} bit in N

(a) If i^{th} bit is already set, leave it set

(b) If i^{th} bit is unset, make it set

$$N = N | (1 \ll i)$$

<< with XOR

$$N = 45 \quad \begin{array}{cccccccc} 2^7 & 2^6 & 2^5 & 2^4 & 2^3 & 2^2 & 2^1 & 2^0 \\ \underline{0} & \underline{0} & \underline{1} & \underline{0} & \underline{1} & \underline{1} & \underline{0} & \underline{1} \end{array}$$

XOR

$$(1 \ll 2) = \begin{array}{cccccccc} \underline{0} & \underline{0} & \underline{0} & \underline{0} & \underline{0} & \underline{1} & \underline{0} & \underline{0} \end{array}$$

$$N^{\wedge} (1 \ll 2) = \begin{array}{cccccccc} \underline{0} & \underline{0} & \underline{1} & \underline{0} & \underline{1} & \underline{0} & \underline{0} & \underline{1} \end{array}$$

$$N = 45 \quad \begin{array}{cccccccc} 2^7 & 2^6 & 2^5 & 2^4 & 2^3 & 2^2 & 2^1 & 2^0 \\ \underline{0} & \underline{0} & \underline{1} & \underline{0} & \underline{1} & \underline{1} & \underline{0} & \underline{1} \end{array}$$

XOR

$$(1 \ll 4) = \begin{array}{cccccccc} \underline{0} & \underline{0} & \underline{0} & \underline{1} & \underline{0} & \underline{0} & \underline{0} & \underline{0} \end{array}$$

$$N^{\wedge} (1 \ll 4) = \begin{array}{cccccccc} \underline{0} & \underline{0} & \underline{1} & \underline{1} & \underline{1} & \underline{1} & \underline{0} & \underline{1} \end{array}$$

$$N = 42 \quad \begin{array}{cccccccc} 2^7 & 2^6 & 2^5 & 2^4 & 2^3 & 2^2 & 2^1 & 2^0 \\ \underline{0} & \underline{0} & \underline{1} & \underline{0} & \underline{1} & \underline{0} & \underline{1} & \underline{0} \end{array}$$

XOR

$$(1 \ll 3) = \begin{array}{cccccccc} \underline{0} & \underline{0} & \underline{0} & \underline{0} & \underline{1} & \underline{0} & \underline{0} & \underline{0} \end{array}$$

$$N^{\wedge} (1 \ll 3) = \begin{array}{cccccccc} \underline{0} & \underline{0} & \underline{1} & \underline{0} & \underline{0} & \underline{0} & \underline{1} & \underline{0} \end{array}$$

$$N = 50 \quad \begin{array}{cccccccc} 2^7 & 2^6 & 2^5 & 2^4 & 2^3 & 2^2 & 2^1 & 2^0 \\ \underline{0} & \underline{0} & \underline{1} & \underline{1} & \underline{0} & \underline{0} & \underline{1} & \underline{0} \end{array}$$

XOR

$$(1 \ll 3) = \begin{array}{cccccccc} \underline{0} & \underline{0} & \underline{0} & \underline{0} & \underline{1} & \underline{0} & \underline{0} & \underline{0} \end{array}$$

$$N^{\wedge} (1 \ll 3) = \begin{array}{cccccccc} \underline{0} & \underline{0} & \underline{1} & \underline{1} & \underline{1} & \underline{0} & \underline{1} & \underline{0} \end{array}$$

Obs: $N^{\wedge} (1 \ll i)$ $\left\{ \begin{array}{l} \text{if } i^{\text{th}} \text{ bit is set, it will be 0} \\ \text{if } i^{\text{th}} \text{ bit is unset, it will be 1} \end{array} \right\} \begin{array}{l} 1 \leftrightarrow 0 \\ \text{Toggle} \end{array}$

Q: Flip i^{th} bit in N

(a) If i^{th} bit is already set, make it unset

(b) If i^{th} bit is unset, set it

$$N = N \oplus (1 \ll i)$$

10:10 → 10:20 PM

* -ve numbers

8bits: $\begin{array}{c} -2^7 \\ 2^6 \quad 2^5 \quad 2^4 \quad 2^3 \quad 2^2 \quad 2^1 \quad 2^0 \\ \hline 1 \quad 1 \quad 1 \quad 1 \quad 1 \quad 1 \quad 1 \end{array}$

-128

$$\hookrightarrow 2^6 + 2^5 + 2^4 + 2^3 + 2^2 + 2^1 + 2^0 = 127$$

Most significant bit: Value for MSB is signed bit

$$\begin{array}{c} -2^7 \quad 2^6 \quad 2^5 \quad 2^4 \quad 2^3 \quad 2^2 \quad 2^1 \quad 2^0 \\ \hline 0 \quad 0 \quad 0 \quad 0 \quad 1 \quad 0 \quad 1 \quad 0 \end{array}$$

$$a: 10 = \underline{0} \quad \underline{0} \quad \underline{0} \quad \underline{0} \quad \underline{1} \quad \underline{0} \quad \underline{1} \quad \underline{0}$$

{ For 32 bits → 31st bit is going to tell the +ve or -ve no.

$$\begin{array}{c} -2^7 \quad 2^6 \quad 2^5 \quad 2^4 \quad 2^3 \quad 2^2 \quad 2^1 \quad 2^0 \\ \hline 1 \quad 0 \quad 0 \quad 0 \quad 1 \quad 0 \quad 1 \quad 0 \end{array} \quad \times$$

$$-10: \underline{1} \quad \underline{0} \quad \underline{0} \quad \underline{0} \quad \underline{1} \quad \underline{0} \quad \underline{1} \quad \underline{0}$$

$$2^1 + 1 + 2^3 + 1 + (-2^7) + 1 = 2 + 8 - 128 = -118$$

Correct way = $-a = \sim a + 1$

$$\begin{array}{c} -2^7 \quad 2^6 \quad 2^5 \quad 2^4 \quad 2^3 \quad 2^2 \quad 2^1 \quad 2^0 \\ \hline 0 \quad 0 \quad 0 \quad 0 \quad 1 \quad 0 \quad 1 \quad 0 \end{array}$$

$$a: 10 = \underline{0} \quad \underline{0} \quad \underline{0} \quad \underline{0} \quad \underline{1} \quad \underline{0} \quad \underline{1} \quad \underline{0}$$

9

1 " 0 0 0 0 0 0 0 1

$$\begin{pmatrix} 1 \\ -2^1 \end{pmatrix} \begin{pmatrix} 1 \\ 2^1 \end{pmatrix} \begin{pmatrix} 1 \\ 2^5 \end{pmatrix} \begin{pmatrix} 1 \\ 2^4 \end{pmatrix} \begin{pmatrix} 0 \\ 2^3 \end{pmatrix} \begin{pmatrix} 1 \\ 2^2 \end{pmatrix} \begin{pmatrix} 1 \\ 2^1 \end{pmatrix} \begin{pmatrix} 0 \\ 2^0 \end{pmatrix}$$

$$= -128 + 64 + 32 + 16 + 4 + 2$$

$$= -128 + 118 = -10$$

$$a = 25 \quad \begin{array}{cccccccc} -2^7 & 2^6 & 2^5 & 2^4 & 2^3 & 2^2 & 2^1 & 2^0 \\ \hline 0 & 0 & 0 & 1 & 1 & 0 & 0 & 1 \end{array}$$

[illegible]

1 1 1 0 0 1 1 1

$$-2^7 \quad 2^6 \quad 2^5 \quad 2^4 \quad 2^3 \quad 2^2 \quad 2^1 \quad 2^0$$

$$-2^7 + 2^6 + 2^5 + 2^2 + 2^1 + 2^0$$

$$= -128 + 64 + 32 + 4 + 2 + 1$$

$$= -128 + 103 = -25$$

long

=

-2^{63}												
63	—	—	—	—	—	—	—	—	—	—	—	—
	.	.	.	.	.	.	7	3	2	1	0	

Ranges

$$8 \text{ bit} = \begin{array}{cccccccc} -2^7 & 2^6 & 2^5 & 2^4 & 2^3 & 2^2 & 2^1 & 2^0 \\ \underline{1} & \underline{0} & \underline{0} & \underline{0} & \underline{0} & \underline{0} & \underline{0} & \underline{0} \end{array} = -2^7 (\text{min})$$

$$\begin{array}{cccccccc} -2^7 & 2^6 & 2^5 & 2^4 & 2^3 & 2^2 & 2^1 & 2^0 \\ \underline{0} & \underline{1} & \underline{1} & \underline{1} & \underline{1} & \underline{1} & \underline{1} & \underline{1} \end{array} = 127 \\ = 2^7 - 1$$

$$\text{Range} = -128 \text{ to } 127 \\ -2^7 \text{ to } 2^7 - 1$$

$$16 \text{ bit} = \begin{array}{cccccccccccc} -2^{15} & & & & & & & & 2^2 & 2^1 & 2^0 \\ \underline{1} & \underline{0} & \underline{0} & \underline{0} & \underline{0} & \underline{0} & \underline{0} & \underline{0} & \underline{0} & \underline{0} & \underline{0} \end{array} = -2^{15}$$

$$\underline{0} \quad \underline{1} \quad \underline{1} \quad \underline{1} \quad \underline{1} \quad \underline{1} \quad \underline{1} \quad \underline{1} \quad \underline{1} = 2^{15} - 1$$

$$\text{Range} = -2^{15} \text{ to } 2^{15} - 1$$

$$\underline{\underline{32 \text{ bit}}} = -2^{31} \text{ to } 2^{31} - 1$$

$$= (-2 * 10^9) \text{ to } (2 * 10^9 - 1) \approx -10^9 \text{ to } 10^9$$

$$2^{10} = 1024 \approx 1000 = 10^3$$

$$2^{30} = 2^{10} * 2^{10} * 2^{10}$$

$$= 10^3 * 10^3 * 10^3$$

$$= 10^9$$

$$2 * 2^{30} = 2 * 10^9$$

— α — α — α — α —

Modular Arithmetic $\rightarrow$ $10^9 + 7$